

Mini Lesson Sample Script

Materials: Personal white board

1. Subtract fractions with like units using a number line.

Solve $5/6 - 4/6$ using a number line.



Answer: $1/6$

T: We know 5 apples minus 4 apples is 1 apple.
Say the subtraction sentence if the unit is sixths.

S: 5 sixths minus 4 sixths equals 1 sixth.

T: Let's show that on a number line. Your number line has endpoints 0 and 1. Partition it into sixths.

S: (Partitions into 6 equal parts.)

T: Plot a point at $5/6$. Now we need to subtract 4 sixths, so we count back by sixths. Move your pencil back with me.

S: 1 sixth, 2 sixths, 3 sixths, 4 sixths.

T: Where did you land?

S: On $1/6$.

T: Draw an arrow above the number line to show subtracting $4/6$. Then write the subtraction sentence in standard form.

	S: $5/6 - 4/6 = 1/6$.
2. Add fractions with like units using a number line.	
Solve $5/4 + 2/4$ using a number line.  <i>Answer: $7/4$</i>	T: We are adding $5/4$ and $2/4$. How should we label the endpoints of the number line? S: $5/4$ is already more than 1, so we need endpoints 0 and 2. T: Yes. Partition each whole into fourths. S: (Partitions into fourths.) T: Plot a point at $5/4$. Then count forward by fourths to add $2/4$. S: 1 fourth, 2 fourths. I landed on $7/4$. T: Draw an arrow above the number line to show adding $2/4$. Say the sum in unit form. S: 5 fourths plus 2 fourths is 7 fourths. T: Write that as an equation using fractions. S: $5/4 + 2/4 = 7/4$.
3. Use a number bond to rename a fraction greater than 1 as a mixed number.	

Use a number bond to decompose $\frac{7}{4}$ into a whole and a fractional part. Then rename $\frac{7}{4}$ as a mixed number.

Answer: $\frac{7}{4} = \frac{4}{4}$ and $\frac{3}{4}$; $\frac{7}{4} = 1 \frac{3}{4}$

T: $\frac{7}{4}$ is greater than 1, so we can rename it as a mixed number. To do that, we decompose it into a whole and a fractional part using a number bond. How many fourths are in 1 whole?

S: 4 fourths.

T: So one part of our number bond is $\frac{4}{4}$. Write $\frac{7}{4}$ as the whole, and put $\frac{4}{4}$ as one of the parts. How many fourths are left?

S: 3 fourths.

T: Record $\frac{3}{4}$ as the other part. $\frac{4}{4}$ is 1 whole, and $\frac{3}{4}$ is the remaining part. So how can we rename $\frac{7}{4}$ as a mixed number?

S: 1 and $\frac{3}{4}$.

T: Right, $\frac{7}{4}$ is the same as $1 \frac{3}{4}$.